THOMAS CHAMBON

AI Data Scientist Computer Vision & GenAI

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RESUME

AI Data Scientist (5+ years of experience) specializing in Computer Vision, NLP, and Deep Learning. Heavily invested in Generative AI, I develop advanced personal projects focused on AI Agents (LangChain, MCP server). My background bridges the gap between scientific rigor (Ph.D., international publications) and industrialization, with proven experience deploying complex models on HPC infrastructures and scaling applications to 1,000+ active users.

SKILLS

- **AI & Deep Learning:** PyTorch, TensorFlow, Hugging Face, Scikit-learn, OpenCV, Spacy, Gensim.
- **Generative AI:** LangChain, MCP server, LLM, RAG.
- **Big Data & Visualization:** Dataiku, Matplotlib, Seaborn, Tableau.
- **Deployment & DevOps:** SLURM, Apptainer, Docker.
- **Databases:** PostgreSQL, MongoDB, Neo4j, Chroma DB.

RELEVANT WORK EXPERIENCE

Research Scientist, R&D – Image Segmentation Optimization

L3i Laboratory, La Rochelle, 2023–2024

- Implemented deep learning models **(Mask2Former, DeepLabV3, EfficientSAM)** for image segmentation (Python).
- Reduced processing time by **50 %** through innovative graph clustering techniques.
- Built large-scale image processing pipelines with **PyTorch, OpenCV, and Scikit-learn** on High-Performance Computing infrastructure **(NVIDIA A100 GPUs, SLURM, Apptainer).**
- Validated models on **over one million images** from benchmark datasets **(COCO, ImageNet)**

Research Scientist, R&D – Analysis of Information Use and Impact on Urban Mobility

L3i Laboratory, La Rochelle, 2019–2023

- Designed and developed an IoT-based mobile tracking application for urban mobility, used by **1,000+ users** with a **90 % retention rate** over 3 months **(Android/iOS).**
- Developed the **data backend**: NoSQL data modeling **(MongoDB)** and REST API for real-time collection and analysis of geolocated data **(~20 million GPS points)** in Python.
- Published scientific papers and presented research at international conferences.
- Supervised and mentored Master's and PhD students in software quality **(metrics, design patterns),** agile project management **(Scrum),** and full data analysis lifecycle **(SQL/NoSQL modeling, data visualization).**

AI Software Developer

CGI, Niort, 2018–2019

- Developed **proof-of-concept (PoC)** solutions for **facial detection and tracking (Computer Vision).** Built conversational assistant prototypes leveraging **NLP** techniques to improve intent recognition and response accuracy.

PROJECTS

- **Cardmaster:** Designed an AI Agent (LLM) for card game deck-building assistance, integrated with RAG **(Hugging Face, LangChain, ChromaDB, Streamlit)**
- **BDthèque:** Collection management tool for books and comics (Python, SQL, Streamlit).
- **Data Protect:** Chromium extension designed to automatically decline Google cookies (JavaScript).

EDUCATION - CERTIFICATIONS

- La Rochelle University: **Ph.D. in Computer Science 2019-2023 | MS.C. Big Data 2017-2019**
- **LangChain Python | Numérique responsable (Green IT) | SecNumacadémie (ANSSI)**

LANGUAGES

- **French** - Native **| English -** Advanced **| German** - Beginner